Analysis #1

1 Part 1: Alignment Between Stereotypes and Movie Preferences

The goal of this problem is to better understand the target audience. Getting a better idea is necessary for improved marketing and development. Although movie ratings provide a benchmark for a movie's success, it is unclear whether certain demographics have distinct preferences. I analyzed the IMDb data to find out if age and gender impact preferences. My initial thought towards this proposed question was that certain genres would show a disproportionate favor with some demographics. By combining both metadata and ratings, I viewed the aggregated rating values by gender for genres and looked at patterns among the 4 unique age groups.

Results are shown in Figure 1. The rating scale ranges from 1 to 10 points. Highlighted in black, the genres, Crime and Romance, show the biggest difference. Males rate Crime movies approximately 0.14 points lower on average (6.38 v. 6.52). Females rate Romance movies 0.25 points higher on average (6.31 v. 6.55). Although the direction of these differences match up, it is still small compared to the scale range. As females tend to rate movies higher, this hints a difference in rating behavior rather than big genre specific preferences. Overall, the minimal difference size suggests that gender is not a dominant factor in explaining the movie rating variance. Directors and marketers that develop strategies based only on gender may gain minimal benefit.

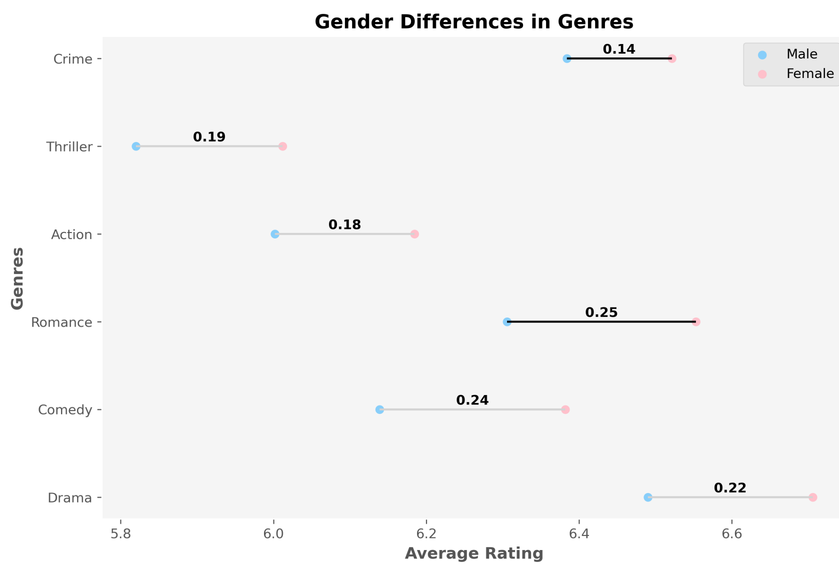

Figure 1: Dumbell plot comparing average ratings between males (blue dots) and females (pink dots) across different movie genres. The horizontal line connecting the genders together shows the difference between the average votes. Romance and Crime are highlighted to show the highest and lowest difference among all 6 genres. Females tend to rate higher across all shown genres.

Results are shown in Figure 2. which displays average movie ratings across genres for different age groups. Drama, Crime, and Romance receive higher ratings across all viewers. This shows a consistent pattern among all age groups for genre preferences. Also, younger audiences rate movies higher than older across all genres. For Drama movies, the average rating difference is 0.69 between the ¡18 and 45+ age groups. The overall ratings decline with increasing age, suggesting a difference in rating behavior. Directors and marketers that develop strategies based only on age may gain minimal benefit. Instead, they should try targeting specific genres to gain more traction.

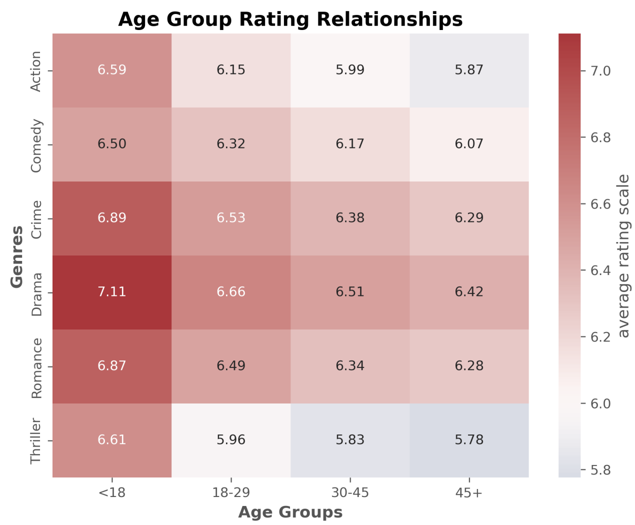

Figure 2: Heatmap showing the distribution of ratings across age groups (< 18, 18–29, 30–45, 45+) for the top six most favored genres. The overall average ratings ranges from 5.78 to 7.11. The given scales marks red for higher values (7) and blue for lower values (5).

2 Part 2: Commercial Success behind Production Budget

The goal of this problem is to better understand how genre preference changes overtime and whether a movie's success is dependent on budget. Although movie ratings can be used, finding these relationships can help developers create a more sound strategy for future content. I analyzed the variance between budgets and average ratings. Also, I observed the overall trend line of six genres across time (1900-2020).

Results are shown in Figure 3. Budget analysis shows high variance among lower budgets. The overall median rating value trends upwards as budget increases. This pattern suggests that while low budget movies can gain both low and high ratings, higher budgeted movies is associated with higher ratings consistently. This could indicate that increasing budgeting allows for higher production quality. However, as there exists a great number of highly rated low budgeted movies, we cannot account finances as the only factor influencing success. Overall, this implies that budget can increase the chances of success, but other factors are needed to drive ratings up.

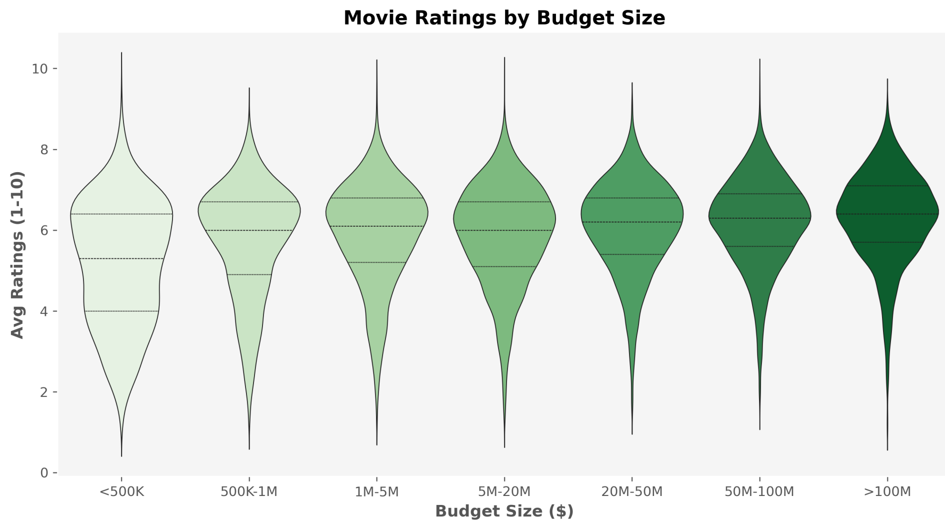

Figure 3: Violin plot showing data density and summary statistics (median, IQR, whiskers) by budget level. Budgets are grouped into seven categories: < \$500K, \$500K–\$1M, \$1M–\$5M, \$5M–\$20M, \$20M–\$50M, \$50M–\$100M, and > \$100M. Ratings range from 1 to 10.

Results are shown in Figure 4. Each graph depicts the trends of genres across time. The movies released by genres is smoothed and linear regression shows the correlations. Action and Thriller trend positively most with year (r=0.85 v. 0.98). Drama and Romance have negative trends (r=-0.60 v. -0.57). Although Drama trends downwards and Thriller goes upwards, Drama is still the most popular genre all across the data. As some genres are more frequent in some time periods, age differences may be influenced by peers or what was available then, not just by personal preference.

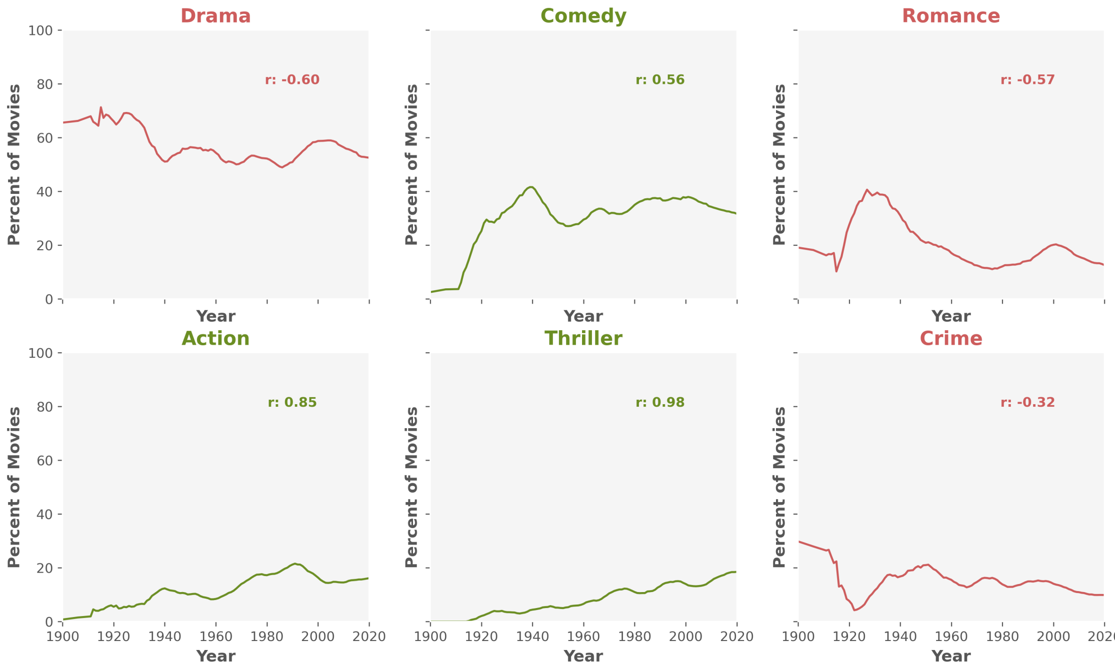

Figure 4: Small multiples plot showing smoothed trends in movies released by genre over time. The shared axes are the smoothed yearly percentage of movies with a genre. By linear regression, green lines indicate a positive trend. Red lines indicate a negative trend.

3 Part 3: Released Seasonal Genre Timeline Pattern

A movie performance can be dependent on the release timing. However, it is unknown how specific genres can relate and be affected by seasons. I wanted to look into if holiday months yielded greater traction. To test this, I looked into the distribution of movies released within the four seasons.
Results are shown in Figure 5. In the summer months, Thriller, Crime and Action (17%, 14%, 14%) held the highest ratings across all seasons for its genre. Romance and Comedy had the highest average ratings in the winter (37% v. 17%). These findings show that genre performance is not the same across the year. Action filled genres tend to perform better in the summer. Relationship focused genres perform better in the winter. This behavior is associated with the sunny, energized vibe that summer brings, whereas holidays in the winter such as Christmas and New Years bring a closeness energy. This suggests that timing does matter. Disregarding seasonal releases may result in missed opportunities to increase success. Developers should consider how seasons match up with genres for better traction.

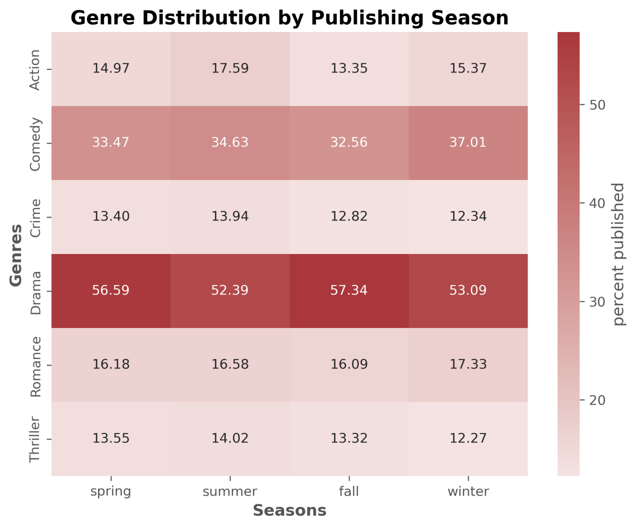

Figure 5: Heatmap showing the percentage of published genres in each season. More concentrated red shows more released movies within that time. Fainter colors suggest fewer releases.

4 Resources used to achieve this goal

I used SeaBorn, Matplotlib, Pandas, and Numpy API Reference Sheet for visualizations and data filtering/cleaning. More specifically, I read manpages on how to annotate graphs and found color references. I was unfamiliar with the calls and arguments for graphs including small multiples and violin plots. I looked into Numpy and Pandas when needed during data selection for each visualization and hypothesis testing. Waskom (2013); Hunter (2003); McKinney (2009)

References

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Analysis #2

1 Part 1: Age and Gender Effects on Action Type Movies

The initial approach I took started out by the question whether gender and age both work together when shaping the ratings for action movies. In other words, does the difference between female and male ratings depend on age? From the age group, I expected younger viewers to rate action movies higher. From the gender group, I expected males to rate action movies higher. For both groups, I expected young males to rate higher among all others. I defined the dependent variable as the average ratings for action like movies (1-10). The two independent variables were gender (female v. male) and age (young: 0-29, old: 30+). The null hypothesis was that gender and age had no interaction, with the alternate being that a relationship existed. To test this, I calculated the weighted averages across the 4 groups and visualized the relationships through lines/scatter plots and bar charts. Also, I computed a paired t test to further confirm my findings.

Results are shown in Figure 1. The parallel slopes show the relationship between average ratings for both genders from young to old, with females above the males. As both head in the same direction, age and gender can be seen as independent factors on rating variance. For marketing and content strategy, this means both demographic factors help and not necessarily scale each other.

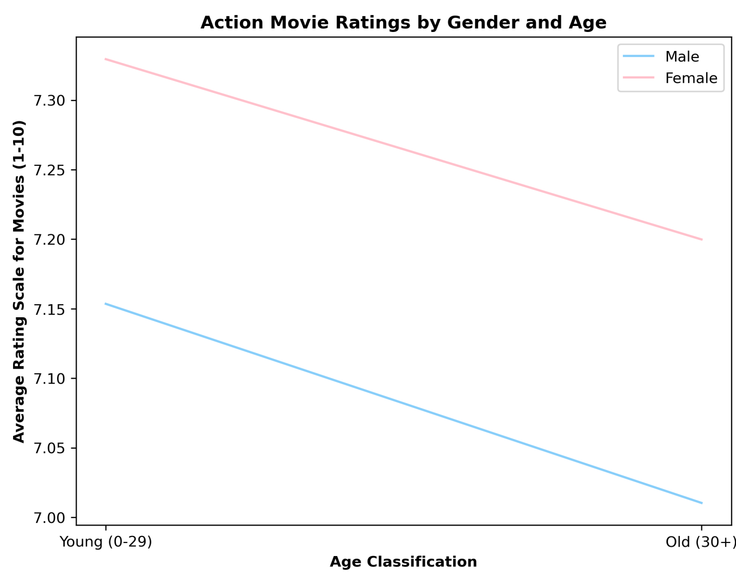

Figure 1: Line plot showing average action movie ratings by gender and age group. The blue line represents males. The pink line represents females. The young group is categorized by combining average ratings from groups ¡18 and 18-29. The old group is combined from 30-45 and 45+.

Results are shown in Figure 2. The scatter plot shows that most movies cluster near the line. This shows that age effects are very similar between females and males as the variation is minimal. We can assume that the impact of one does not change the effect of the other. Again, the same conclusion can be made for developers.

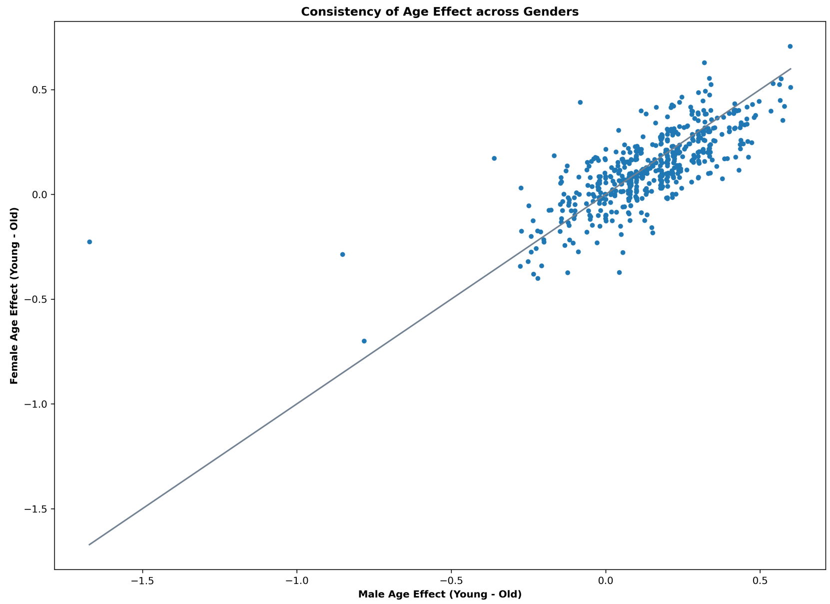

Figure 2: Scatter plot compares the effect of age on ratings for males (x) and females (y) across movies. Each point represents a movie. The position shows how much the younger viewers rate it higher than older viewers. Points near the line show that age effect is almost the same for both genders.

Results are shown in Figure 3. The bar chart shows the paired rating differences across age and gender. Male rate action movies around 0.18 lower than females in both age groups. Younger viewers rate action movies .14 points higher than the old for both genders. These differences are statistically significant. This shows both age and gender independently influence ratings. For developers, this suggests that both factors should be considered when observing their target's preferences.

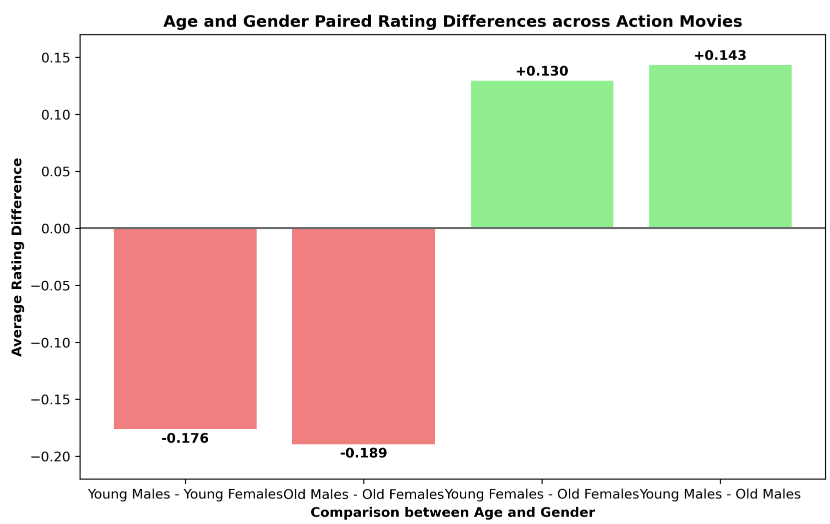

Figure 3: Bar chart showing paired rating difference across age and gender. Red bars signal that the difference is negative. Green bars signal that the difference is positive. Values above the bars show the difference between the demographics while holding one constant.

2 Part 2: Independent Effects of Age and Gender on Action Type Movie Ratings

There are two goals to this problem. The first is to find out if younger viewers rate action movies significantly higher than older viewers. The second is to find out if female viewers rate action movies significantly higher. Again the independent variables are defined the same. The dependent, movie ratings, filters for content with high action scenes. This includes the genres Thriller, Action, Adventure, and War. The null hypotheses are that there are no significant differences between action movie ratings with age and gender independently. The alternates are the goals stated previously. To test this, I calculated weighted average ratings and did a paired t test to compare young v. old and female v. male for each action movie.

Results are shown in Figure 4. The paired t test shows a highly significant age effect (t=20.67, p = 6.29e-71), allowing us to reject the null hypothesis. We are very confident that younger viewers rate action movies an average of 0.16 points higher than older viewers. This can show that younger audiences prefer more action level content. Another take could be that younger and older audiences have different rating styles. This 0.16 takes up 1.6% on a 10 point scale. Developers should take age into account, but also know that there are many more predictors that play into movie success.

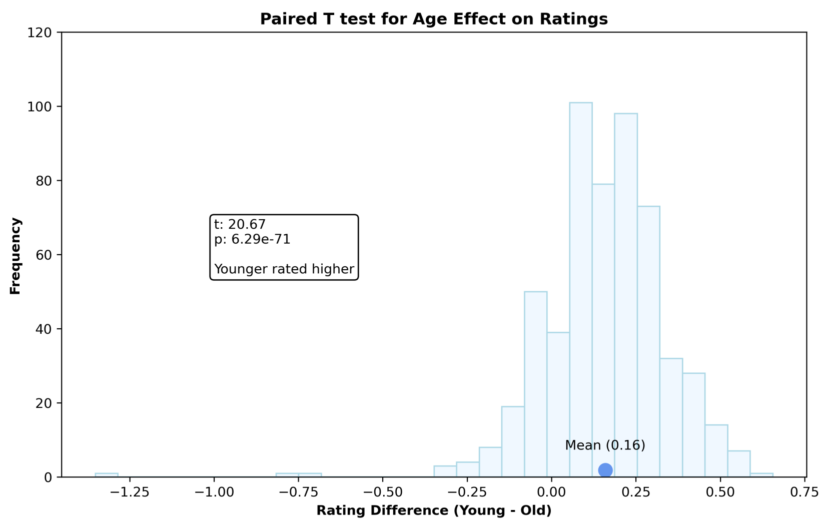

Figure 4: Paired t test histogram showing the distribution of young and old rating differences across action level movies. The x axis represents how much higher younger viewers rated. The y axis shows the number of movies with each difference. The blue dot is placed were the mean difference of 0.16 is. The box displays the two tail t and p value.

Results are shown in Figure 5. The paired t test shows a highly significant gender effect (t=13.82, p = 1.45e-37), allowing us to reject the null hypothesis. We are very confident that female viewers rate action movies an average of 0.20 points higher than male viewers. This can show that female audiences prefer more action level content. Another take could be that female and male audiences have different rating styles. This 0.20 takes up 2.0% on a 10 point scale. Developers should take gender into account, but also know that there are many more predictors that play into movie success.

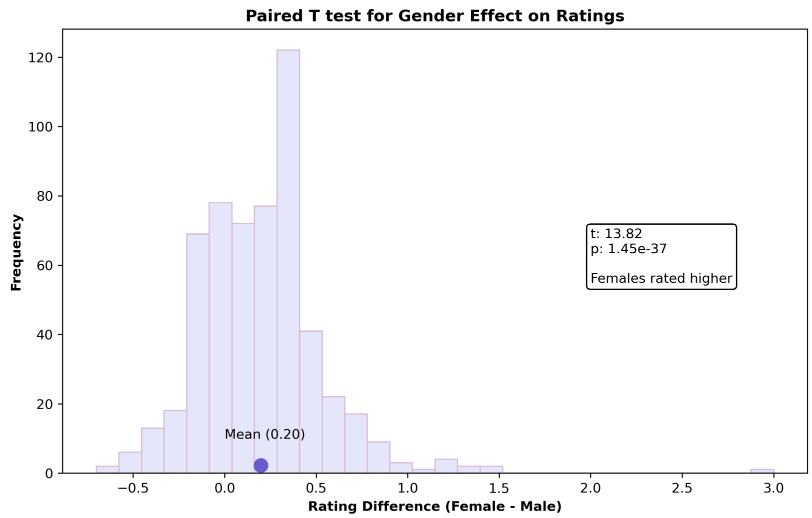

Figure 5: Paired t test histogram showing the distribution of female and male rating differences across action level movies. The x axis represents how much higher female viewers rated. The y axis shows the number of movies with each difference. The purple dot is placed were the mean difference of 0.20 is. The box displays the two tail t and p value.

3 Resources used to achieve this goal

I used SeaBorn, Matplotlib, Pandas, and Numpy API Reference Sheet for visualizations and data filtering/cleaning. More specifically, I read manpages on how to annotate graphs and found color references. I was unfamiliar with the calls and arguments for graphs including small multiples and violin plots. I looked into Numpy and Pandas when needed during data selection for each visualization and hypothesis testing. Waskom (2013); Hunter (2003); McKinney (2009)

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Analysis #3

1 Part 1: Budget Influence Over Movie Profits

The goal of this was to find whether movie budgets had an effect on it's success. On top of that, another question was if this relationship held differently across varying genres. In the EDA, I found that higher budgets had higher ratings, but the correlation between another success factor, income, and budget wasn't addressed. Also, different genres might benefit more from higher budgets. For this, I selected Drama and Action for the genres. The budget was defined as high and low where the median budget value was the decision boundary. The question was if increasing budget had the same effect on gross income for Drama and Action movies. To test this, I filtered the data and calculated the mean across all groups to visualize the relationship better.

Results are shown in Figure 1. The line plot shows a big difference between genres and budget type for success. For Drama movies, the budget increase caused the average gross income to increase by $48M. For Action movies, the budget increased as well with a $158M difference. Between Action and Drama, the difference is $109M with Action benefitting more due to its steep slope. Action movies may need more financial backup due to the stunts and special effects involved. Overall, this suggests that higher budgeting decreases the risk of movie failure.

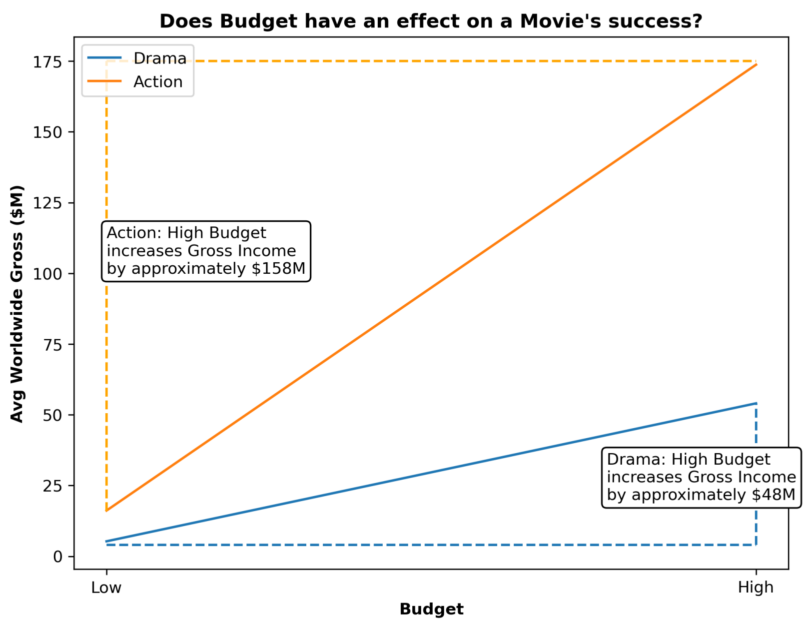

Figure 1: Line plot showing average worldwide gross income ($M) by budget type for Drama (blue) and Action (orange) movies. The x axis shows the budget type (high v. low). The y axis shows the average gross income. The annotations shows the increase difference from low to high budget

2 Part 2: Observing Profit Data Behavior

The goal of this is to look into the distribution of gross incomes across each group to determine if the variance has an effect on our data interpretation. For continuous values, especially across a large range, outliers can skew the data heavily. The huge mean differences found in Part 1 could be misleading. To test this, I created boxplots to see the distributions across (Drama, Low), (Drama, High), (Action, Low), and (Action, High). This was done on the original income and scaled income through log. I also found the ratio of mean to media revenue for each group as larger means hint skewness.

Results are shown in Figure 2 and 3. Figure 2 shows two boxplots. The left is scaled and the right is the original distribution. The scaled income plot shows a more symmetric spread, while the original shows highly right skewness. This confirms that the gross income is not normally distributed. Figure 3 quantifies this skewness through the ratio between mean and median. Drama Low have an extreme skewness with a factor of 12.1. This means that the average gross income was over 12 times larger than the median because of outliers. As all groups display some level of skewness, the data needs to be transformed for our next step (ANOVA).

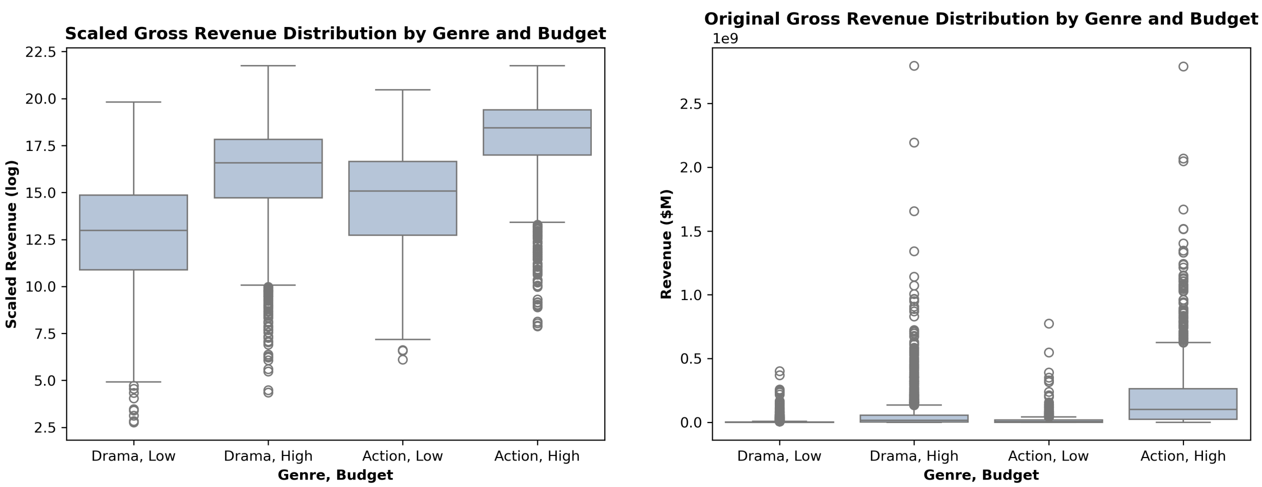

Figure 2: Two boxplots comparing scaled (log) and original worldwide gross income distributions across 4 groups. The dots represent the outliers. The center line represents the median. The box represent the IQR where 50% of the data lies. The whiskers represent the boundaries of normal values (1.5*IQR)

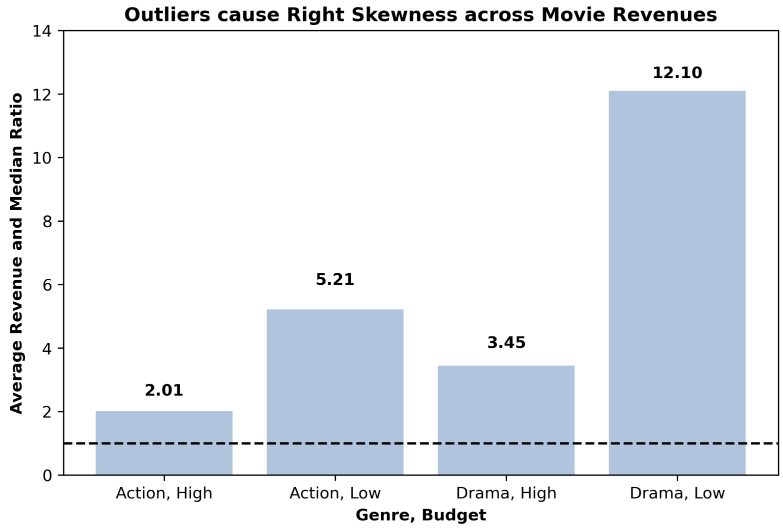

Figure 3: Bar chart showing the mean to median factor for each group. The dashed line where y=1 indicates where mean = median. Ratios (numbers above the bars) greater than one indicate right skewed distributions.

3 Part 3: Relation Findings between Genre and Budgeting over Revenue

The goal of this is to complete a formal hypothesis testing to find out the relationships between genre and budget with movie revenue. While Part 1 showed that there is a great difference, we need to use ANOVA and the scaled gross income to test if the pattern is statistically significant. I created three null hypotheses:

1. Genre has no significant effect on revenue
2. Budget has no significant effect on revenue
3. There is no significant interaction between genre and budget on revenue

To test these hypotheses, I do a two way ANOVA test on the scaled incomes (dependent variable). The independent variables were budget (low v. high) and genre (drama v. action). I used Type 2 sum of squares to observe these three relationships.

Results are shown in Figure 4. The table shows highly significant effects from both genre ($F = 635$, $p = 1.83e\text{-}135$) and budget ($F=3512$, $p < 0.001$). This allows us to reject the null. From here we can conclude that Action and Drama movie revenues differ greatly. Also, higher budgets create more revenue than low budget across both genres. The interaction term ($F=0.00019$, $p=0.989$) was too small, so we fail to reject the null. This contradicts our findings in Part 1. This indicates that the outliers created the wrong interpretation. Instead, both genres show around the same increase when moving from low to high budget when on a log scale. This suggests that budget boosts revenue regardless of genre. Overall, genre mostly changes how much money a movie starts with, not how much more money it gains from having a bigger budget.

Success Relations between Genre, Revenue, and Budget

	sum_sq	df	F	PR(>F)
C(genre)	4219.5781	1.0	635.3793	0.0
C(budget)	23324.108	1.0	3512.1178	0.0
C(genre):C(budget)	0.0013	1.0	0.0002	0.989
Residual	58906.0068	8870.0	nan	nan

Figure 4: ANOVA Table showing the mean results across genre types and budget levels. Large differences between groups mean strong effects. This is confirmed by high F and low p values. The highlighted p values show how statistically significant the findings were (red: not v. green: apparent)

4 Resources used to achieve this goal

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I referenced on page on Geeks for Geeks to understand how to perform a 2 way ANOVA in Python. GeeksforGeeks (2025)

References

GeeksforGeeks (2025). How to perform a two-way anova in python. `https://www.geeksforgeeks.org/machine-learning/how-to-perform-a-two-way-anova-in-python/`.

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